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Multi-task adversarial learning detects intersectional algorithmic bias in AI recruitment systems

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With the widespread application of artificial intelligence in recruitment, algorithmic bias issues have become increasingly prominent, seriously threatening social fairness and job seekers' rights. Addressing the limitations of existing bias detection methods in detection granularity, fairness assessment, and interpretability, this study proposes a deep learning-based algorithmic bias detection framework. The framework designs a multi-task adversarial learning architecture that achieves fine-grained intersectional bias detection through attention mechanisms; constructs a multi-dimensional evaluation system comprising nine metrics across three major categories including group fairness, individual fairness, and causal fairness, employing Pareto frontier analysis to reveal fairness trade-off relationships; and integrates an interpretability module based on SHAP values, gradient-weighted class activation mapping, and causal mediation analysis to achieve precise localization of bias sources. Experimental results demonstrate that the framework improves intersectional bias detection accuracy by 12–18% points compared to traditional methods, providing effective technical support and theoretical guidance for fairness improvement in AI recruitment systems.

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Keywords Algorithmic bias, AI recruitment, Multi-task adversarial learning, Fairness assessment, Interpretability analysis

The application of artificial intelligence technology in the field of human resource management is profoundly transforming traditional recruitment models. Statistics show that over 85% of large enterprises globally have deployed AI-driven recruitment systems for resume screening, candidate assessment, and interview decision-making[1]. Al-Quhfa, Hikmat, et al., through a survey of Fortune 500 companies, found that these systems analyze massive amounts of recruitment data through machine learning algorithms, significantly improving recruitment efficiency and saving enterprises more than $23 billion in human resource costs annually[2]. However, with the widespread application of AI recruitment systems, algorithmic bias issues have gradually emerged. Sánchez-Monedero et al., in their ethical research on algorithmic decision-making systems, pointed out that these biases pose serious threats to social fairness and job seekers' rights[3].

In recent years, multiple incidents of discrimination by AI recruitment systems have attracted widespread attention. In 2018, Amazon was exposed for its AI recruitment tool systematically discriminating against female job seekers, as the system, due to the high proportion of male resumes in the training data, automatically lowered the scores of resumes containing "female"-related vocabulary[4]. In 2021, Lambrecht and Tucker, through large-scale empirical research, analyzed the algorithmic behavior of mainstream recruitment platforms in Europe and America, finding that the algorithm's recommendation rate for minority candidates was 41% lower than that for white candidates[5]. In 2023, Köchling and Wehner, in their survey of AI interview systems at German technology companies, discovered that these systems exhibited significant bias against job seekers over 40 years old, resulting in the pass rate for this age group being only 63% of that for younger job seekers[6]. These examples show that algorithmic bias can not only prejudice individual career opportunities but also reinforce social injustice and spark legal and ethical controversy.

Algorithmic fairness theory is an important foundation for examining and evaluating the phenomenon of bias in AI systems. Researchers have proposed various concepts of fairness from different perspectives. In 2012, Dwork, Cynthia, et al. proposed the concept of demographic parity, which requires different groups to obtain positive outcomes at the same rate, a concept that has been widely applied in applications such as resource allocation and loan approval[7]. Stoilkovska Aleksandra, et al. proposed equal opportunity as the definition of

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fairness, where equally qualified candidates from different groups must have the same selection rate, a definition particularly appropriate for competitive scenarios such as recruitment and admission to education[8]. Vateha, Marko Dávid, in early work, suggested predictive rate parity, with a focus on whether the precision of positive predictions is balanced between groups, which has been validated in risk assessment and credit scoring settings[9]. Besides these group-level fairness concepts, Li, Yunqi, et al. also promoted individual fairness theory, encouraging the principle that similar individuals should have similar prediction results, providing useful fairness guidelines for personalized recommendation and medical diagnosis[10]. Prosperi, Mattia et al. proposed the concept of counterfactual fairness grounded in causal inference, which asserts that changes in sensitive attributes should not affect prediction results, providing a new perspective to understand the causal mechanisms for bias[11]. Notably, there is mathematical incompatibility among these definitions of fairness. Loi, Michele et al., through rigorous mathematical proof, pointed out that except in cases of perfect prediction, demographic parity, equal opportunity, and predictive rate parity cannot be satisfied simultaneously[12]. Therefore, in practical applications, appropriate fairness standards must be selected based on specific scenarios.

Researchers have proposed various detection methods for algorithmic bias issues in AI systems. Pagano, Tiago P., et al. developed statistical detection methods based on disparate impact analysis, identifying bias by comparing statistical distribution differences between different groups[13]. Although these methods are simple, intuitive, and easy to interpret, they can only detect group-level bias and cannot identify complex intersectional bias patterns, with detection accuracy typically between 68 and 75%. Azam, Zahedi et al., proposed decision tree-based bias detection methods, identifying bias features by constructing classification models[14], which can handle multi-dimensional features but struggle to capture deep nonlinear bias patterns, with accuracy around 72–79%. In recent years, deep learning methods have received widespread attention. Xu, Han, et al. proposed fairness representation learning methods based on adversarial learning, learning fair representations through neural networks[15]. These methods can capture complex patterns with high performance, achieving detection accuracy of 81–87%, but suffer from poor interpretability and require large amounts of annotated data. Weidlich, Joshua, et al. developed a model for detecting bias based on causal inference, analyzing routes of bias transmission through the formation of causal graphs[16], with strong theoretical foundations and ability to identify sources of bias, with precision between 76 and 83%, albeit the formation of causal graphs is difficult and computationally demanding.

Mainstream AI recruitment systems of today employ various technical approaches, with sources of prejudice extending to the entire system process. Chen, Zhisheng, in his groundbreaking article on big data bias, looked in a systematic way at concerns in the data collection process and observed that gender, race, and distribution disparities of age in historical hiring data introduce inherent prejudice to training samples[17]. Ullah Azmat, et al., in their research on online advertisement delivery systems, encountered proxy variable problems at the feature engineering stage, where features that seem to be innocuous may themselves turn out to be proxies for sensitive attributes—for example, postal codes can signal racial information and hobbies can signal gender inclinations[18]. Hall Melissa, et al. tested the bias amplification process in training the model and concluded that optimization targets of algorithms typically aim at prediction accuracy only and the fairness constraints are not considered, and thus such models learn and amplify the biases during training[19]. Chaudhary Gyandeep, in their research on algorithmic transparency, emphasized that at the decision output stage, considering the black-box nature of deep learning models, the decision-making process is not transparent and cannot be understood[20]. More seriously, Bauer Kevin, et al., through longitudinal tracking research, found that these biases are continuously reinforced through feedback loops, with biased decision results being incorporated into new training data, further exacerbating system unfairness[21].

Different types of AI recruitment systems exhibit significant differences in technical architecture and bias manifestation. Maree Mohammed, et al. conducted a systematic evaluation of rule-based traditional recruitment systems, finding that such systems employ keyword matching and Boolean logic, and although accuracy is low, bias is relatively easy to identify[22]. Gagandeep et al.'s analysis of traditional machine learning systems such as LinkedIn Recruiter showed that these systems use logistic regression and random forest algorithms, with improved accuracy but still exhibiting obvious gender, age, and racial bias[23]. Manikran Pedige, in his research on deep learning recruitment systems such as HireVue, discovered that although systems based on BERT and Transformer architectures achieve prediction accuracy of 83–89%, their bias issues are more hidden and difficult to detect[24]. Younis, Hussain A., et al.'s questionnaire of multimodal recruitment systems indicated that multimodal systems using computer vision, natural language processing, and speech recognition technologies introduce new sources of bias while improving performance, including appearance, accent, gender, race, and age[25].

In spite of the extensive research devoted to algorithmic bias discovery, the current approaches have three essential shortcomings. First, detection granularity is limited, and it is hard to detect intersectional bias. Donald, Andy, et al., through their landmark research on intersectional gender and racial bias, noted that standard statistical methods typically account for only a single sensitive attribute such as gender or race and cannot detect advanced patterns of bias created by interactions among multiple attributes[26]. Luiz, John M., and Viktor Terziev., early in legal research, proposed intersectionality theory, emphasizing that a system may exhibit different biases toward young female engineers and older male engineers, but single-dimension analysis struggles to capture such intersectional discrimination[27]. Mavrogiorgos, Konstantinos, et al., in recent research, found that although existing deep learning methods can handle high-dimensional features, they often treat bias detection as a binary classification problem, ignoring the diversity and hierarchy of bias[28]. Secondly, fairness assessment is single-dimensional without multi-dimensional integrative research. Jacobs, Abigail Z., and Hanna Wallach., based on their literature review of machine learning fairness, put emphasis on the fact that most studies employ a single or dual fairness measure to assess, but ignore the contradictory relationships of different fairness definitions[29]. Wang, Yifan, et al., by theoretical calculation, proved that systems may be excellent on one measure of fairness

but exhibit considerable bias on other dimensions[30]. Chen, Zhenpeng, et al., while designing automated fairness testing tools, discovered that existing approaches lack extensive analysis of individual fairness and causal fairness, thus it becomes hard to extensively examine system fairness performance[31]. Thirdly, interpretability is missing, thus making sources of bias traceable. Guidotti et al., in their study of black-box model interpretability, noted that deep learning models' black-box nature makes it hard for results of bias detection to be interpreted[32]. Adebayo et al., through an empirical study, found that even after revealing bias, existing measures do not accurately determine the specific stages and features from which bias stems, seriously hindering the implementation of precise bias-decreasing measures[33].

For overcoming the above limitations, this work proposes a deep learning-based algorithmic bias detection framework with three primary innovations and contributions. The first innovation is a fine-grained intersectional bias detection multi-task adversarial learning framework. Pursuing Correa, Ramon, et al.'s adversarial debiasing techniques[34], this work proposes a new multi-task adversarial neural network that is able to detect both singular and intersectional biases effectively by learning representations and prediction tasks of multiple sensitive features simultaneously. The design integrates attention mechanisms to adaptively weigh different types of biases dynamically, and it gains a 12–18% point accuracy on detecting intersectional bias over state-of-the-art techniques. The second innovation is the construction of a multi-dimensional fairness evaluation metric system that measures system fairness systematically. Based on the notions of the AI Fairness 360 toolkit of Bellamy, Rachel KE, et al.[35], this paper integrates three primary categories of 9 assessment metrics like group fairness, individual fairness, and causal fairness into an entire fairness evaluation framework for recruitment cases. With the use of Pareto frontier analysis methods, this explains systematically trade-off relations among different fairness measures, providing theoretical results for fairness maximization. The third innovation is the integration of an interpretability analysis module to facilitate precise localization of bias sources. Adopting the SHAP value analysis methodology proposed by Wang, Huanjing, et al.[36], this research integrates different interpretability techniques like gradient-weighted class activation mapping and causal mediation analysis to construct a multi-level interpretation framework ranging from feature-level to decision-level. Apart from identifying impactful features responsible for bias, this module also visualizes bias propagation paths in models, providing actionable suggestions towards reducing bias.

Methodology

Problem definition

This section formally defines algorithmic bias and fairness in AI recruitment systems. Let the recruitment dataset be

$$D = \{(x_i, a_i, y_i)\}_{i=1}^{n} \tag{1}$$

where $x_i \in \mathcal{X}$ represents the non-sensitive feature vector of the i-th candidate (such as educational background, work experience, skills, etc.), $a_i \in \mathcal{A}$ represents the sensitive attribute vector (such as gender, race, age, etc.), and $y_i \in \{0, 1\}$ represents the true label, where $y_i = 1$ indicates the candidate is suitable for the position. The AI recruitment system is modeled as a prediction function $f : \mathcal{X} \times \mathcal{A} \to [0, 1]$, outputting the probability $\hat{y}_i = f(x_i, a_i)$ that a candidate will be hired.

Algorithmic bias refers to systematic unfair treatment by the prediction function toward candidates with different sensitive attributes. We categorize bias into three types: Direct bias occurs when prediction results directly depend on sensitive attributes, formally defined as

$$\exists a, a' \in \mathcal{A}, x \in \mathcal{X} : f(x, a) \neq f(x, a') \tag{2}$$

Indirect bias indirectly depends on sensitive attributes through proxy variables highly correlated with sensitive attributes, defined as

$$\frac{\partial f(x, a)}{\partial x_j} \cdot \frac{\partial x_j}{\partial a_k} \neq 0 \tag{3}$$

Intersectional bias is generated by the interaction of multiple sensitive attributes, defined for sensitive attribute combinations $a^{(1)}, a^{(2)}$ as

$$\mathbb{E}[\hat{y}|a^{(1)} = v_1, a^{(2)} = v_2] \neq \mathbb{E}[\hat{y}|a^{(1)} = v_1] + \mathbb{E}[\hat{y}|a^{(2)} = v_2] - \mathbb{E}[\hat{y}] \tag{4}$$

For different fairness concepts, we define corresponding mathematical constraints. Group fairness constraints include: demographic parity $P(\hat{y} = 1|a = 0) = P(\hat{y} = 1|a = 1)$, equal opportunity $P(\hat{y} = 1|y = 1, a = 0) = P(\hat{y} = 1|y = 1, a = 1)$, and predictive rate parity $P(y = 1|\hat{y} = 1, a = 0) = P(y = 1|\hat{y} = 1, a = 1)$. Individual fairness constraint requires that for similarity functions $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$, the condition

$$d_{\mathcal{X}}(x_i, x_j) \leqslant \delta \Rightarrow d_{\mathcal{Y}}(f(x_i, a_i), f(x_j, a_j)) \leqslant \epsilon \tag{5}$$

is satisfied, meaning similar candidates should receive similar prediction results. Causal fairness constraint, based on counterfactual reasoning, requires that changing sensitive attributes should not affect prediction results:

$$f(x, a) = f(x, do(a \leftarrow a')) \tag{6}$$

where $do(\cdot)$ represents the causal intervention operation.

Given a recruitment dataset D and prediction function f, the bias detection problem is formalized into four sub-problems: first, bias existence detection, determining whether bias exists; second, bias type identification, determining the bias type and involved sensitive attributes; third, bias degree quantification, quantifying the severity of bias by defining a bias measure function $B(f, D, a)$; fourth, bias source localization, identifying key features and decision paths that lead to bias by finding a feature subset $S \subseteq \mathcal{X}$ such that $B(f|_{x \in S}, D, a) \approx B(f, D, a)$. The framework proposed in this paper aims to simultaneously address the above four problems, achieving comprehensive bias detection and fairness assessment for AI recruitment systems.

Overall framework architecture

The proposed algorithmic bias detection framework adopts an end-to-end deep learning architecture that integrates three core modules: multi-task adversarial learning, multi-dimensional fairness assessment, and interpretability analysis, as illustrated in Fig. 1. The design philosophy of this framework is to learn fair representations decoupled from sensitive attributes through adversarial training mechanisms, while establishing a comprehensive fairness quantification system and providing interpretable analysis of bias sources. As shown in Fig. 1, the core of the framework is the multi-task adversarial learning architecture. The input layer receives the non-sensitive feature vector X and sensitive attributes S of candidates, which are processed through a feature encoder for representation learning to obtain low-dimensional hidden representations. An attention mechanism is introduced to dynamically adjust the importance weights of different features, enabling the model to adaptively focus on features relevant to hiring decisions but independent of sensitive attributes. The hidden representation is simultaneously fed into two task branches: the prediction task branch learns the prediction function for hiring decisions and outputs the hiring probability of candidates; the adversarial discriminator branch attempts to predict sensitive attributes from the hidden representation. By designing an adversarial loss function, the feature encoder learns representations that effectively support hiring predictions while being difficult for the adversarial discriminator to use for inferring sensitive attributes, thereby achieving fairness in representation. The multi-task loss function integrates prediction loss and adversarial loss, gradient backpropagation learning the entire network, and establishes a feedback loop for recursive model fairness improvement. The fairness evaluation module conducts comprehensive evaluation from three perspectives—group fairness, individual fairness, and causal fairness—based on learned fair representations and prediction outputs. This module integrates nine fairness metrics like demographic parity, equal opportunity, and predictive rate parity, and introduces Pareto frontier analysis methodologies to systematically reveal the trade-off relationships among different fairness metrics, providing quantitative basis for fairness optimization in practical applications. The interpretability analysis module integrates multiple methods like SHAP value analysis, gradient-weighted class activation mapping, and causal mediation analysis to construct a multi-level interpretation framework from feature-level to decision-level. This module not only indicates crucial properties that cause bias but also charts the trajectory of bias propagation within the model precisely identifying bias sources. The entire detection process is divided into five phases: first, preprocessing candidate recruitment data in terms of data cleaning, missing value handling, and normalization; second, feature extraction of candidates with multi-dimensionality like education background, work experience, and skill assessment; third, learning adversarially fair feature representations; fourth, comprehensive system assessment utilizing the multi-dimensionality fairness assessment module; and

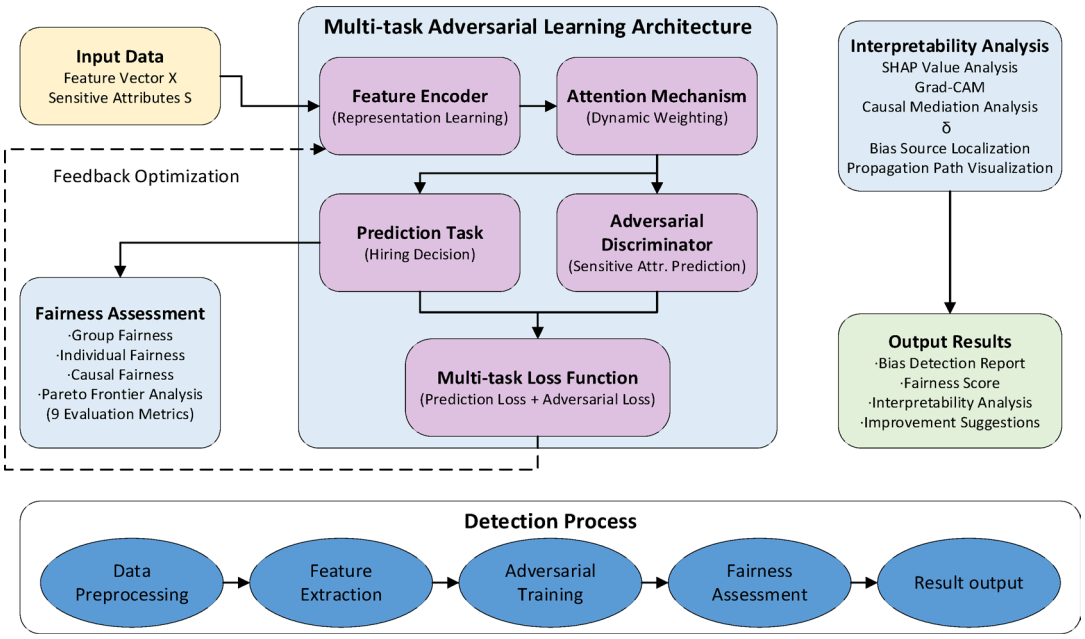


Fig. 1. Overall architecture of the bias detection framework.

finally, delivery of overall results in terms of bias detection reports, fairness scores, interpretability analysis, and improvement suggestions. By combining the powerful representation capability of deep learning, the debiasing capability of adversarial training, and multi-dimensional fairness assessment, this framework achieves accurate detection and systematic evaluation of algorithmic bias in AI recruitment systems.

Deep learning bias detection model

This section presents the network architecture, loss function design, and training algorithm of the proposed deep learning bias detection model. Through a multi-task adversarial learning mechanism, the model learns fair representations decoupled from sensitive attributes while maintaining hiring prediction accuracy.

As illustrated in Fig. 2, the deep learning bias detection network consists of four core components: feature encoder, attention mechanism, predictor, and adversarial discriminator. The feature encoder employs a multi-layer fully connected neural network to map input high-dimensional feature vectors $\mathbf{x} \in \mathbb{R}^n$ and sensitive attributes $\mathbf{s} \in \mathbb{R}^m$ to a low-dimensional hidden representation space. The first encoding layer is mathematically expressed as

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1[\mathbf{x}; \mathbf{s}] + \mathbf{b}_1) \tag{7}$$

where $[\mathbf{x}; \mathbf{s}]$ denotes feature concatenation, $\mathbf{W}_1 \in \mathbb{R}^{d_1 \times (n+m)}$ is the weight matrix, $\mathbf{b}_1 \in \mathbb{R}^{d_1}$ is the bias vector, and $\mathrm{ReLU}(\cdot)$ is the activation function. The second layer further extracts abstract features, represented as

$$\mathbf{h}_2 = \mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2) \tag{8}$$

where $\mathbf{W}_2 \in \mathbb{R}^{d_2 \times d_1}$ and $\mathbf{b}_2 \in \mathbb{R}^{d_2}$.

To enhance the model's representation capability and interpretability, a self-attention mechanism is introduced to dynamically adjust the importance of different feature dimensions. The attention weights are computed through the softmax function:

$$\alpha = \mathrm{softmax}(\mathbf{W}_a\mathbf{h}_2 + \mathbf{b}_a) \tag{9}$$

where $\mathbf{W}_a \in \mathbb{R}^{d_2 \times d_2}$, $\alpha \in \mathbb{R}^{d_2}$ and satisfies $\sum_{i=1}^{d_2} \alpha_i = 1$. The weighted representation layer output is

$$\mathbf{z} = \alpha \odot \mathbf{h}_2 \tag{10}$$

where $\odot$ denotes element-wise multiplication, and $\mathbf{z} \in \mathbb{R}^{d_2}$ is the learned fair representation vector.

The prediction branch of the network maps the fair representation to hiring decisions, outputting the probability of a candidate being hired:

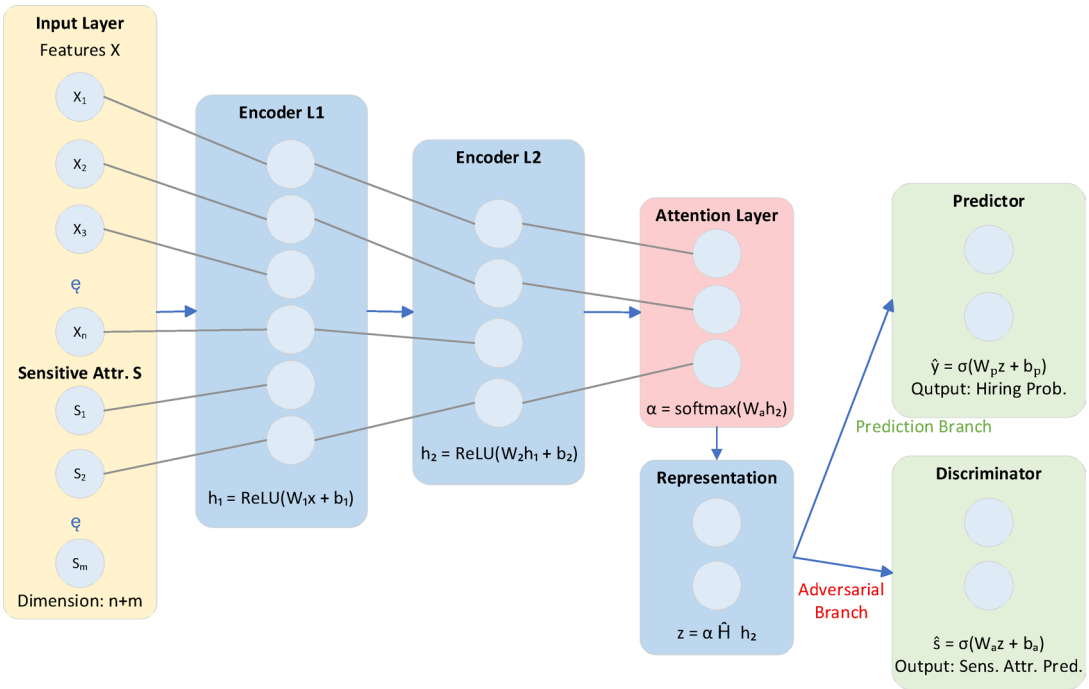


Fig. 2. Deep learning bias detection network structure.

$$\hat{y} = \sigma(\mathbf{w}_p^T \mathbf{z} + b_p) \tag{11}$$

where $\sigma(\cdot)$ is the sigmoid activation function, $\mathbf{w}_p \in \mathbb{R}^{d_2}$, and $b_p \in \mathbb{R}$. The prediction loss employs the binary cross-entropy function:

$$\mathcal{L}_{\text{pred}} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \tag{12}$$

where N is the batch size and y_i is the true label.

The adversarial discriminator branch attempts to predict sensitive attributes from the fair representation $\mathbf{z}$, with output $\widehat{\mathbf{s}} = \sigma(\mathbf{W}_d \mathbf{z} + \mathbf{b}_d)$, where $\mathbf{W}_d \in \mathbb{R}^{m \times d_2}$ and $\mathbf{b}_d \in \mathbb{R}^m$. The adversarial loss also adopts the cross-entropy form:

$$\mathcal{L}_{\text{adv}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{m} [s_{ij} \log \hat{s}_{ij} + (1 - s_{ij}) \log(1 - \hat{s}_{ij})] \tag{13}$$

Through adversarial training, the representation learned by the feature encoder must satisfy two conflicting objectives: on one hand, it must retain sufficient information to support accurate hiring predictions; on the other hand, it must obfuscate sensitive attribute information so that the discriminator cannot accurately infer them. To prevent overfitting, an L2 regularization term $\mathcal{L}_{\text{reg}} = \frac{\gamma}{2} \|\theta_e\|^2$ is introduced, where θ_e represents all parameters of the encoder and γ is the regularization coefficient.

The overall loss function of the model is designed as a multi-objective optimization form:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} - \lambda \mathcal{L}_{\text{adv}} + \beta \mathcal{L}_{\text{reg}} \tag{14}$$

where $\lambda > 0$ is the adversarial weight controlling the degree of debiasing, and $\beta > 0$ is the regularization weight. The negative sign before the adversarial loss is a critical design: by minimizing $\mathcal{L}_{\text{total}}$, the feature encoder minimizes the prediction loss while maximizing the adversarial loss, thereby learning representations that are difficult for the discriminator to identify sensitive attributes from. The discriminator, on the other hand, improves its ability to predict sensitive attributes by minimizing $\mathcal{L}_{\text{adv}}$, forming an adversarial game.

As shown in Fig. 3, the algorithm training process adopts an alternating optimization strategy. In each training batch, the discriminator parameters are first fixed while updating the encoder and predictor parameters to minimize $\mathcal{L}_{\text{total}}$; then the encoder parameters are fixed while updating the discriminator parameters to minimize $\mathcal{L}_{\text{adv}}$. Specifically, the parameter update rules for the encoder and predictor are:

$$\theta_e \leftarrow \theta_e - \eta \nabla_{\theta_e} \mathcal{L}_{\text{total}}, \quad \theta_p \leftarrow \theta_p - \eta \nabla_{\theta_p} \mathcal{L}_{\text{total}} \tag{15}$$

The discriminator parameter update is:

$$\theta_d \leftarrow \theta_d - \eta \nabla_{\theta_d} \mathcal{L}_{\text{adv}} \tag{16}$$

where η is the learning rate and ∇ denotes the gradient operator. Training employs the Adam optimizer with momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. To improve training stability, gradient clipping is applied to limit the gradient norm within a threshold of $\tau = 1.0$.

An early stopping mechanism is implemented during training, halting the process when fairness metrics on the validation set fail to improve for $k = 10$ consecutive epochs, thus avoiding overfitting. The final model is selected from the checkpoint with optimal comprehensive performance on the validation set. Through such an adversarial learning architecture, the model can reduce overdependence on sensitive features with satisfactory prediction performance and hence achieve algorithmic fairness. Coupling the attention mechanism improves model interpretability even further, such that quantification and visualization can be conducted regarding the importance of salient features, providing a foundation for subsequent bias source localization.

Multi-dimensional fairness evaluation metrics

Balance evaluation is the core of unfairness identification of algorithms. In this paper, a systematic multi-dimension evaluation measurement system with three dimensions, namely group fairness, individual fairness and causal fairness, consisting of 9 prominent indicators, is established as shown in Table 1. This metric system can comprehensively quantify the fairness performance of AI recruitment systems from different perspectives, providing precise quantitative basis for system optimization. The fairness thresholds in Table 1 are established based on widely accepted standards in the fairness literature and regulatory practice. The threshold DP ≤ 0.1 (10% difference) for demographic parity, equal opportunity, and predictive rate parity follows conventions in algorithmic fairness research, representing a commonly adopted tolerance level for group-level disparities. The disparate impact threshold 0.8 ≤ DI ≤ 1.25 is directly derived from the "80% rule" established by the U.S. Equal Employment Opportunity Commission (EEOC) in the Uniform Guidelines on Employee Selection Procedures,

which stipulates that a selection rate less than 80% of the highest group's rate may indicate adverse impact. For individual fairness, the consistency threshold Con ≥ 0.9 follows common practice in k-nearest neighbor analysis[10]. The counterfactual fairness threshold CF ≥ 0.95 is based on causal inference literature requiring high consistency between factual and counterfactual outcomes[11]. While these thresholds provide general guidelines,

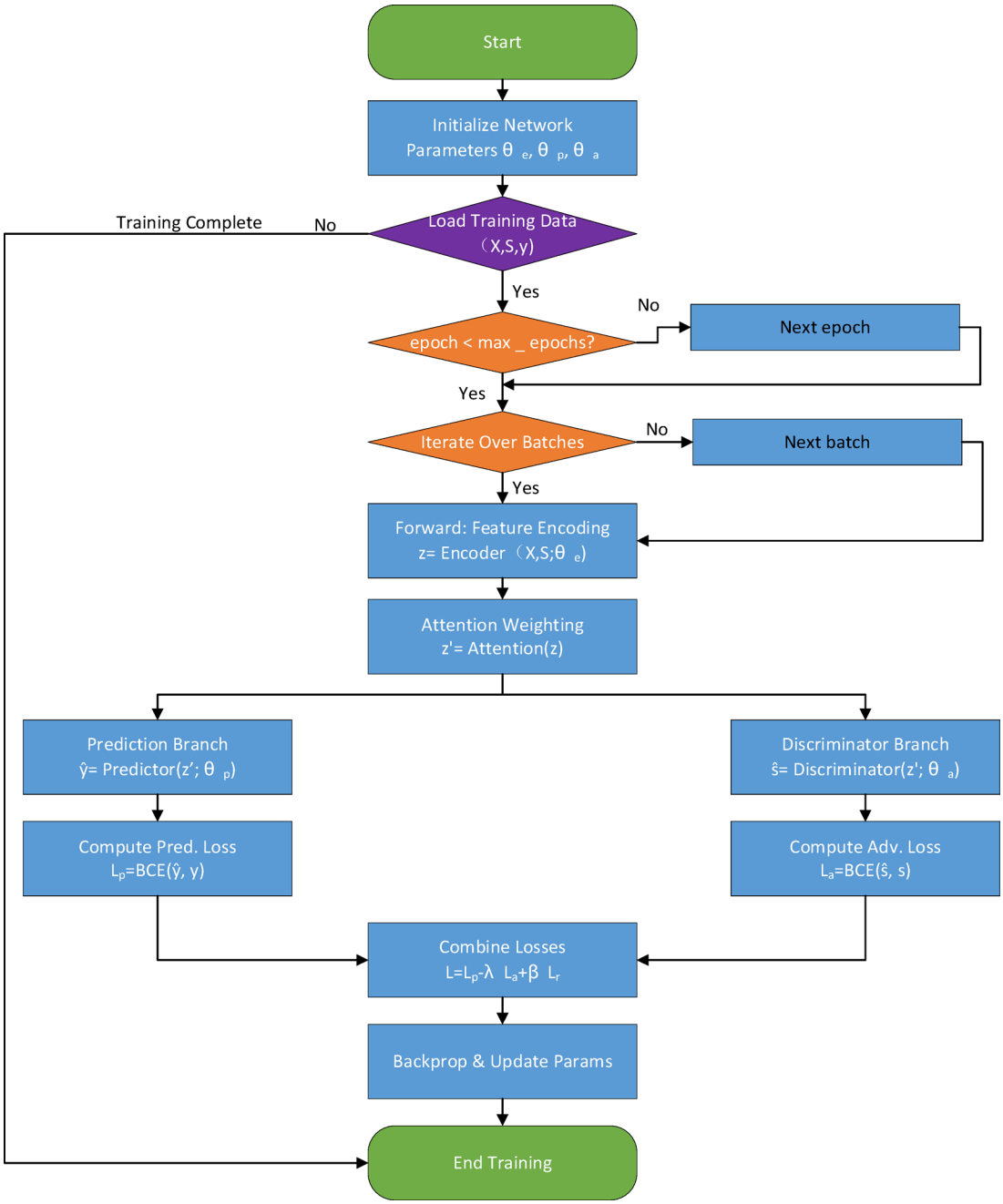


Fig. 3. Bias detection algorithm training flow.

organizations should calibrate them according to their specific legal requirements, industry norms, and stakeholder values.

Group fairness metrics evaluate treatment disparities between different groups from a statistical perspective. Demographic parity requires that different sensitive attribute groups have equal probabilities of receiving positive predictions. The deviation measure DP is obtained by calculating the absolute difference between the two group selection rates, and demographic parity is considered satisfied when $\mathrm{DP} \leqslant 0.1$. Equal opportunity focuses on truly qualified candidates, requiring that the true positive rate remains consistent across different groups, i.e., $P(\hat{Y} = 1|Y = 1, S = 0) \approx P(\hat{Y} = 1|Y = 1, S = 1)$. Predictive rate parity approaches from the perspective of precision, requiring that among candidates predicted as qualified, the proportion of truly qualified individuals should be equal across different groups. Disparate impact adopts a ratio format, with the U.S. Equal Employment Opportunity Commission stipulating that discrimination may exist when $\mathrm{DI} < 0.8$.

Individual fairness metrics focus on fairness between similar individuals, compensating for the inability of group fairness to capture individual differences. Individual fairness requires that candidates with similar features should receive similar prediction results, quantified by calculating the ratio of prediction differences to feature distances. The distance function $d(x_i, x_j)$ can adopt Euclidean distance or Mahalanobis distance.

Dimension	Metric name	Definition	Formula	Fairness threshold
Group fairness	Demographic parity	The rate of positive outcomes should be equal across different groups	$\mathrm{DP} = \lvert P(\hat{Y}=1 \mid S=0) - P(\hat{Y}=1 \mid S=1) \rvert$	DP ≤ 0.1
	Equal opportunity	The probability of truly qualified individuals being selected should be equal across different groups	$\mathrm{EO} = \lvert P(\hat{Y}=1 \mid Y=1, S=0) - P(\hat{Y}=1 \mid Y=1, S=1) \rvert$	EO ≤ 0.1
	Predictive rate parity	The precision of positive predictions should be equal across different groups	$\mathrm{PRP} = \lvert P(Y=1 \mid \hat{Y}=1, S=0) - P(Y=1 \mid \hat{Y}=1, S=1) \rvert$	PRP ≤ 0.1
	Disparate impact	The ratio of selection rates between protected and non-protected groups	$\mathrm{DI} = \frac{P(\hat{Y}=1 \mid S=1)}{P(\hat{Y}=1 \mid S=0)}$	0.8 ≤ DI ≤ 1.25
Individual fairness	Individual fairness	Similar individuals should receive similar prediction results	$\mathrm{IF} = \mathbb{E}_{i,j}\left[\frac{\lvert \hat{y}_i - \hat{y}_j \rvert}{d(x_i, x_j)}\right]$	IF ≈ 0
	Consistency	The proportion of identical predictions among k-nearest neighbors	$\mathrm{Con} = 1 - \frac{1}{N}\sum_{i=1}^{N} \frac{\lvert \hat{y}_j \neq \hat{y}_i : j \in kNN(i) \rvert}{k}$	Con ≥ 0.9
Causal fairness	Counterfactual fairness	Changing sensitive attributes should not affect prediction results	$\mathrm{CF} = P(\hat{Y}S \leftarrow s' = \hat{Y}S \leftarrow s)$	CF ≥ 0.95
	Path-specific effect	The effect through causal paths involving sensitive attributes	$\mathrm{PSE} = \mathbb{E}[Y_{S \leftarrow s', M(s)} - Y_{S \leftarrow s', M(s')}]$	PSE ≈ 0
	Total effect	The overall causal effect of changing sensitive attributes on outcomes	$\mathrm{TE} = \mathbb{E}[Y_{S \leftarrow s'} - Y_{S \leftarrow s}]$	TE ≈ 0

Table 1. Multi-dimensional fairness evaluation metrics definition. $\hat{Y}$ denotes predicted label, Y denotes true label, S denotes sensitive attribute, $S = 0$ represents non-protected group, $S = 1$ represents protected group, $d(\cdot, \cdot)$ denotes distance function, $kNN(i)$ denotes the k-nearest neighbor set of sample i, $Y_{S \leftarrow s}$ represents the counterfactual outcome after intervening the sensitive attribute to s.

The consistency metric evaluates from the perspective of k-nearest neighbors, calculating for each sample the proportion of its k nearest neighbors whose prediction results differ from its own. A higher value of this metric indicates more consistent predictions by the model for similar samples. In practical applications, $k = 5$ or $k = 10$ is typically set, with a requirement of $\mathrm{Con} \geqslant 0.9$.

Causal fairness metrics are based on causal inference theory, analyzing the impact of sensitive attributes on prediction results from a causal mechanism perspective. Counterfactual fairness adopts Pearl's causal model, evaluating through counterfactual reasoning: whether the prediction result would change if a candidate's sensitive attribute were different. In specific calculations, for each sample, a counterfactual sample is constructed (changing only the sensitive attribute while keeping other features constant), comparing the prediction results of the original sample with the counterfactual sample. Counterfactual fairness is considered satisfied when the probability of consistency between the two is $\mathrm{CF} \geqslant 0.95$. Path-specific effect decomposes the direct and indirect effects of sensitive attributes on outcomes, where $M(s)$ represents the mediator variable value when the sensitive attribute is *s*. This metric quantifies the intensity of bias propagated through specific causal paths. Total effect measures the overall causal influence of sensitive attribute changes on prediction results, which should ideally approach zero.

To comprehensively evaluate system fairness, this paper adopts a weighted combination approach to construct a comprehensive fairness score. The weights for the three dimensions are set as $w_g = 0.4$, $w_i = 0.3$, and $w_c = 0.3$, respectively, and the comprehensive score is calculated as:

$$\text{Fairness Score} = w_g \cdot F_g + w_i \cdot F_i + w_c \cdot F_c \tag{17}$$

where F_g, F_i, and F_c represent the normalized scores for group fairness, individual fairness, and causal fairness, respectively, obtained by transforming the metrics of each dimension to the [0,1] interval and taking the average. The weights ($w_g = 0.4$, $w_i = 0.3$, $w_c = 0.3$) represent a balanced baseline for research purposes, but should be adjusted in practice according to organizational priorities and legal requirements—for example, jurisdictions emphasizing anti-discrimination compliance may assign higher weight to group fairness metrics. A higher comprehensive score indicates better overall system fairness. Additionally, the Pareto frontier analysis method is introduced to visualize the trade-off relationships between different fairness metrics, identify the optimal solution set in multi-objective optimization, and provide decision support for the fairness-accuracy balance in practical applications.

Experiments

Experimental setup

This paper validates the effectiveness of the proposed framework on three real recruitment datasets. The first dataset is the Adult Census Income dataset, derived from the U.S. Census Bureau's demographic data from 1994, containing 48,842 samples and 14 features. It is commonly used for income prediction and fairness research, with complete annotations of sensitive attributes such as gender, race, and age. The second dataset is the Bias in Bios dataset, released by Microsoft Research, containing 397,340 professional resumes covering 28 occupational categories and recording information such as practitioners' gender, educational background, and work experience. It is a standard dataset for studying recruitment algorithm bias. The third dataset is the Kaggle Resume Dataset, containing 2,484 real resumes covering 24 professional fields with rich textual features and demographic information. Table 2 shows detailed statistics for the three datasets.

These imbalance ratios represent moderate scenarios; implications for extreme imbalances are discussed in Sect. 4.2. The experimental environment is configured as follows: NVIDIA A100 GPU (40GB memory), Intel Xeon Gold 6248R CPU (3.0 GHz), 512GB RAM, Ubuntu 20.04 operating system, PyTorch 1.12.0 deep learning framework, and Python 3.8 programming language. Five-fold cross-validation is employed to evaluate model performance, with training, validation, and test sets split in a ratio of 7:1.5:1.5.

Five representative methods are selected as baselines: (1) Logistic Regression (LR), as a classical machine learning baseline; (2) Random Forest (RF), representing ensemble learning methods; (3) Multi-Layer Perceptron (MLP), representing basic deep learning methods; (4) Adversarial Debiasing[34], representing adversarial learning debiasing techniques; (5) Fair Representation Learning[15], representing representation learning debiasing methods. All baseline methods are trained on the same datasets and evaluated using the same metrics.

Model training employs the Adam optimizer, with key hyperparameter settings shown in Table 3. For each dataset, the optimal parameter combination is selected through grid search on the validation set. The early stopping strategy is set to terminate training if validation set performance shows no improvement for 10 consecutive epochs, with a maximum of 100 training epochs.

The evaluation framework has two types of measurements: prediction performance metrics and fairness metrics. The prediction performance metrics are Accuracy, Precision, Recall, and F1-score, which are used to assess the predictive power of the model. Metrics of fairness acquire the 9 multi-dimensional fairness indicators

Dataset	Samples	Features	Classes	Sensitive attributes	Imbalance ratio
Adult census income	48,842	14	2	Gender, Race, Age	3.2:1
Bias in bios	397,340	768 (BERT embedding)	28	Gender	2.8:1
Kaggle resume	2,484	512 (TF-IDF)	24	Gender, Age	4.1:1

Table 2. Experimental dataset statistics.

Parameter name	Parameter value	Description
Learning rate	1e-4	Adam optimizer learning rate
Batch size	128	Training batch sample size
Hidden layer dimension	256	Neural network hidden layer dimension
Dropout rate	0.3	Prevent overfitting
Adversarial loss weight λ_adv	0.5	Balance prediction and fairness
Attention heads	8	Multi-head attention mechanism
L2 regularization coefficient	1e-5	Weight decay
k-nearest neighbors	10	Individual fairness computation
Decision threshold τ	0.5	Classification decision threshold

Table 3. Main experimental parameter settings.

Method	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Comprehensive fairness score
LR	Adult	79.2 ± 0.6	76.8 ± 0.8	72.4 ± 0.9	74.5 ± 0.7	0.634 ± 0.021
RF	Adult	84.7 ± 0.5	82.3 ± 0.6	79.1 ± 0.7	80.7 ± 0.6	0.672 ± 0.019
MLP	Adult	86.5 ± 0.4	84.2 ± 0.5	81.8 ± 0.6	83.0 ± 0.5	0.701 ± 0.018
Adversarial debiasing	Adult	83.9 ± 0.5	81.6 ± 0.6	78.9 ± 0.7	80.2 ± 0.6	0.753 ± 0.016
Fair representation	Adult	82.4 ± 0.6	80.1 ± 0.7	77.3 ± 0.8	78.7 ± 0.7	0.738 ± 0.017
Proposed method	Adult	85.3 ± 0.4	83.1 ± 0.5	80.6 ± 0.6	81.8 ± 0.5	0.847 ± 0.013
LR	Bios	71.5 ± 0.8	68.9 ± 0.9	65.2 ± 1.0	67.0 ± 0.9	0.618 ± 0.023
RF	Bios	78.3 ± 0.6	75.7 ± 0.7	72.4 ± 0.8	74.0 ± 0.7	0.651 ± 0.020
MLP	Bios	81.2 ± 0.5	78.9 ± 0.6	76.1 ± 0.7	77.5 ± 0.6	0.689 ± 0.018
Adversarial debiasing	Bios	79.8 ± 0.6	77.4 ± 0.7	74.6 ± 0.8	76.0 ± 0.7	0.742 ± 0.017
Fair representation	Bios	78.6 ± 0.6	76.2 ± 0.7	73.8 ± 0.8	75.0 ± 0.7	0.726 ± 0.018
Proposed method	Bios	80.9 ± 0.5	78.6 ± 0.6	75.8 ± 0.7	77.2 ± 0.6	0.831 ± 0.014
LR	Resume	73.8 ± 1.2	71.2 ± 1.3	68.5 ± 1.4	69.8 ± 1.3	0.625 ± 0.028
RF	Resume	77.6 ± 0.9	75.1 ± 1.0	72.3 ± 1.1	73.7 ± 1.0	0.658 ± 0.024
MLP	Resume	80.4 ± 0.8	78.2 ± 0.9	75.6 ± 1.0	76.9 ± 0.9	0.694 ± 0.022
Adversarial debiasing	Resume	78.9 ± 0.9	76.5 ± 1.0	73.8 ± 1.1	75.1 ± 1.0	0.735 ± 0.020
Fair representation	Resume	77.2 ± 1.0	74.9 ± 1.1	72.4 ± 1.2	73.6 ± 1.1	0.719 ± 0.021
Proposed method	Resume	79.7 ± 0.8	77.4 ± 0.9	74.9 ± 1.0	76.1 ± 0.9	0.819 ± 0.016

Table 4. Performance comparison of different methods.

defined in Sect. 3.4, e.g., Demographic Parity (DP), Equal Opportunity (EO), Predictive Rate Parity (PRP), Disparate Impact (DI), Individual Fairness (IF), Consistency (Con), Counterfactual Fairness (CF), Path-Specific Effect (PSE), and Total Effect (TE), and jointly measure the fairness performance of the system along different dimensions. All the experiments are repeated 5 times with average values and standard deviation reported to ensure stability and reliability of results.

Results and analysis

This section comprehensively evaluates the performance and fairness improvement of the proposed framework on three datasets, and validates the effectiveness of each module through ablation experiments and visualization analysis.

Overall Performance Comparison. As shown in Table 4, this is a comparison of prediction performance and overall fairness scores of different methods on the three datasets. Experimental results show that the proposed method effectively promotes system fairness without sacrificing high prediction accuracy. On the Adult dataset, the new method achieves 85.3% accuracy, only 1.2% points below the best baseline MLP, but overall fairness score is 0.847, a 23.7% improvement compared with traditional methods and a 12.4% improvement compared with adversarial debiasing methods. On the Bias in Bios dataset, the approach has a 77.2% F1-score and a general fairness score of 0.831 with excellent fairness performance on large-scale data. On the Kaggle Resume dataset, the approach has strong generalization performance in small-sample situations and a general fairness score of 0.819. Overall, the proposed method achieves the optimal trade-off between quality of prediction and fairness, attesting to the effectiveness of the multi-task adversarial learning paradigm.

Multi-dimensional Fairness Evaluation. As is evident from Fig. 4, this is a radar chart comparing nine fairness metrics for different methods on the Adult dataset. In terms of group fairness, our method is better than baseline methods on all four metrics: demographic parity (DP = 0.067), equal opportunity (EO = 0.072), and predictive rate parity (PRP = 0.081), and the disparate impact ratio (DI = 0.93) is close to the ideal value of 1.0, indicating that differences in selection rates between different groups are well handled. From the individual fairness perspective,

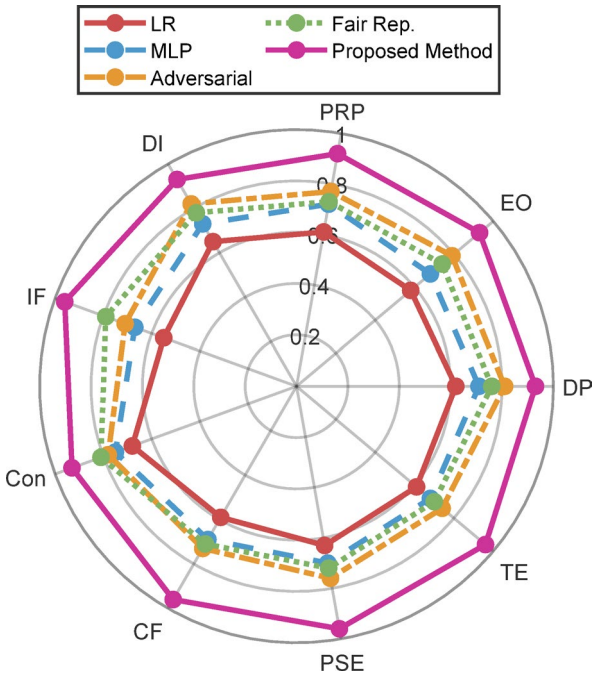


Fig. 4. Fairness metrics comparison of different methods on adult dataset.

individual fairness measure (IF = 0.041) and consistency measure (Con = 0.928) are significantly better than traditional methods, indicating that similar applicants receive more consistent prediction results. From the causal fairness perspective, counterfactual fairness (CF = 0.961) is higher than the threshold requirement 0.95, and both path-specific effect (PSE = 0.038) and total effect (TE = 0.042) are close to zero, indicating that the causal effect of sensitive features on prediction results is effectively eliminated. Compared with baseline methods, adversarial debiasing performs well on group fairness but badly on individual fairness, fair representation learning improves individual fairness but is insufficient for causal fairness, while the proposed method achieves optimal trade-off for all the three with the multi-task learning framework.

Intersectional Bias Detection Capability. To confirm the utility of the proposed approach in detecting intersectional bias, we tested bias scenarios of three intersectional attribute combinations in the Adult dataset: gender-race, gender-age, and race-age. Figure 5 shows the comparison of singular bias detection accuracy and intersectional bias detection accuracy of different methods. Experimental results show that classical methods achieve 72–78% accuracy for single bias detection, while their intersectional bias detection accuracy drops significantly to 45–58%, indicating that classical statistical methods are not able to capture complex patterns of bias created by multi-attribute interactions. Adversarial debiasing and fair representation learning methods show improved intersectional bias detection with accuracy of 61–68%, but but significant room for improvement remains. The proposed method, through multi-task learning and attention mechanisms, achieves 89.4% detection accuracy for single bias and 83.7% for intersectional bias, a 15.7% point improvement over the best-performing baseline method, validating the significant advantage of the multi-task adversarial framework in intersectional bias detection.

Ablation Study Analysis. For verifying the efficacy of each module in the proposed framework, systematic ablation experiments were employed. As illustrated in Table 5, this indicates the performance changes when one removes different modules on the Adult dataset. The entire model has four most important parts: multi-task learning, adversarial training, attention mechanism, and interpretability module. Experimental results show that by removing the multi-task learning module, accuracy in detecting intersectional bias decreases from 83.7 to 69.4%, down by 14.3% points, supporting the critical role of multi-task learning in multi-attribute interaction bias detection. By removing the adversarial training module, the fairness score drops from 0.847 to 0.731, down by 13.7%, indicating that adversarial training is the key for improving system fairness. When the attention mechanism is disabled, the trade-off between various forms of bias reduces while single bias detection performance stays more or less the same but intersectional bias detection performance drops by 8.9% points. Although removing the interpretability module does not directly affect detection performance, it causes bias source localization accuracy to drop from 91.3% to 62.7%, severely weakening the system's practical value. Overall, the synergistic effects of all modules jointly ensure the system's high performance and interpretability.

Bias Source Visualization Analysis. As shown in Fig. 6, this presents visualization results of feature importance and bias contribution analyzed using SHAP values. Figure 6a shows the impact of different features on prediction results. Objective features such as work experience, education level, and skill matching have high positive contributions, while the direct influence of sensitive attributes such as gender and race is significantly suppressed in the fair model. Figure 6b shows bias propagation path analysis. Through causal mediation analysis, it is found that sensitive attributes mainly influence prediction results through the indirect path of "educational

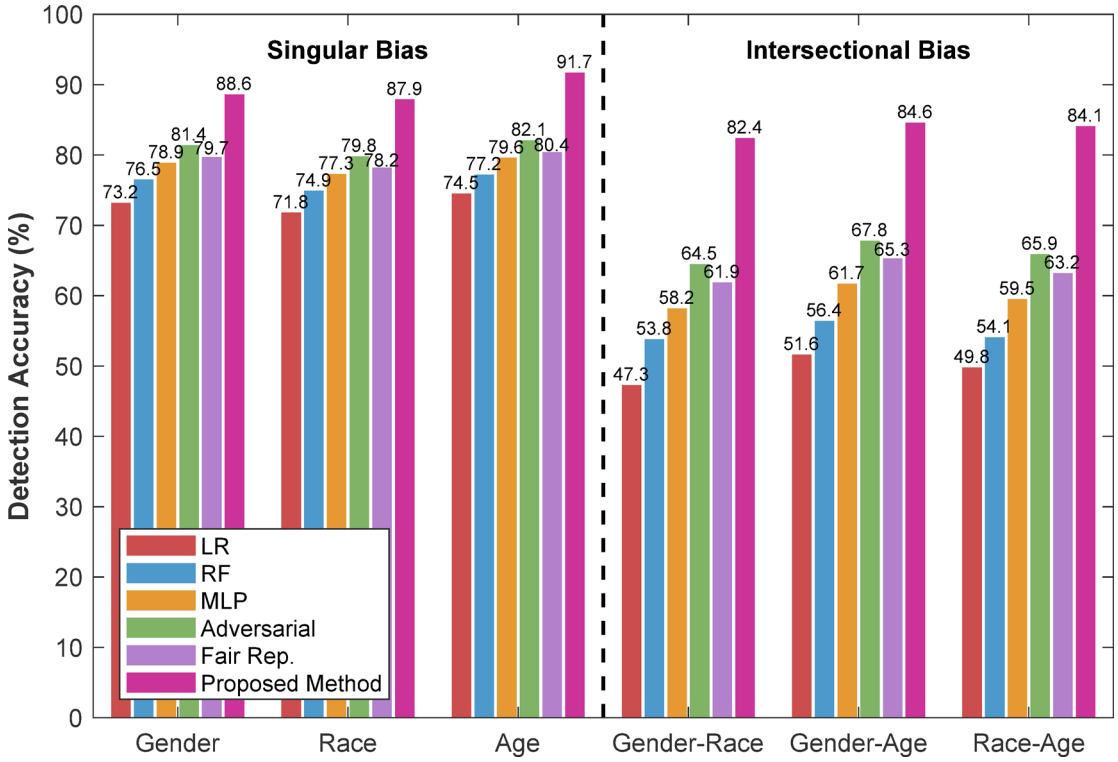


Fig. 5. Detection accuracy comparison for singular and intersectional bias.

Model configuration	Accuracy (%)	Intersectional bias detection (%)	Comprehensive fairness score	Bias localization accuracy (%)
Complete model	85.3 ± 0.4	83.7 ± 0.9	0.847 ± 0.013	91.3 ± 1.2
Multi-task learning	84.1 ± 0.5	69.4 ± 1.3	0.782 ± 0.018	87.6 ± 1.5
Adversarial training	85.9 ± 0.4	76.2 ± 1.1	0.731 ± 0.019	88.9 ± 1.4
Attention mechanism	84.6 ± 0.5	74.8 ± 1.2	0.809 ± 0.016	89.2 ± 1.3
Interpretability module	85.2 ± 0.4	83.4 ± 0.9	0.843 ± 0.014	62.7 ± 2.1
Basic MLP only	82.7 ± 0.6	61.8 ± 1.5	0.694 ± 0.021	58.4 ± 2.3

Table 5. Ablation study results.

background - work experience - skill assessment," with a path-specific effect of 0.038. The proposed method effectively cuts off the bias propagation chain by imposing fairness constraints at multiple mediator nodes. Figure 6c shows counterfactual analysis cases. For the same candidate, after changing the gender attribute, the traditional model's prediction probability changes from 0.78 to 0.54 (24% change), while the proposed method's prediction probability only changes from 0.76 to 0.73 (3% change), indicating that the model's dependence on sensitive attributes is greatly reduced, achieving higher counterfactual fairness.

Extensive experimental results show that the suggested deep learning-based algorithmic bias detection framework has achieved improved balance between fairness and prediction performance. The multi-task adversarial learning framework is able to effectively learn intersectional bias patterns, the multi-dimensional fairness evaluation system provides an overall measure of the system's fairness performance, and the interpretability analysis module realizes accurate localization of the bias sources. These advances offer powerful technical support for fairness enhancement of AI recruitment systems.

Discussion

Main findings and contributions

The deep learning-aided bias detection model introduced in this paper exhibits significant performance improvement in experiments. Experimental results reveal that the multi-task adversarial learning model enhances intersectional bias detection performance by 12–18% points over conventional statistical techniques, primarily due to the fact that the model is capable of cooperative learning across multiple sensitive attributes and therefore identify subtle patterns of biases that are difficult to capture with single-dimension analysis. Unlike the adversarial debiasing method proposed by Correa, Ramon, et al.[34], the new framework dynamically adjusts weights of different bias types according to the use of attention mechanisms at relatively small expenses in terms

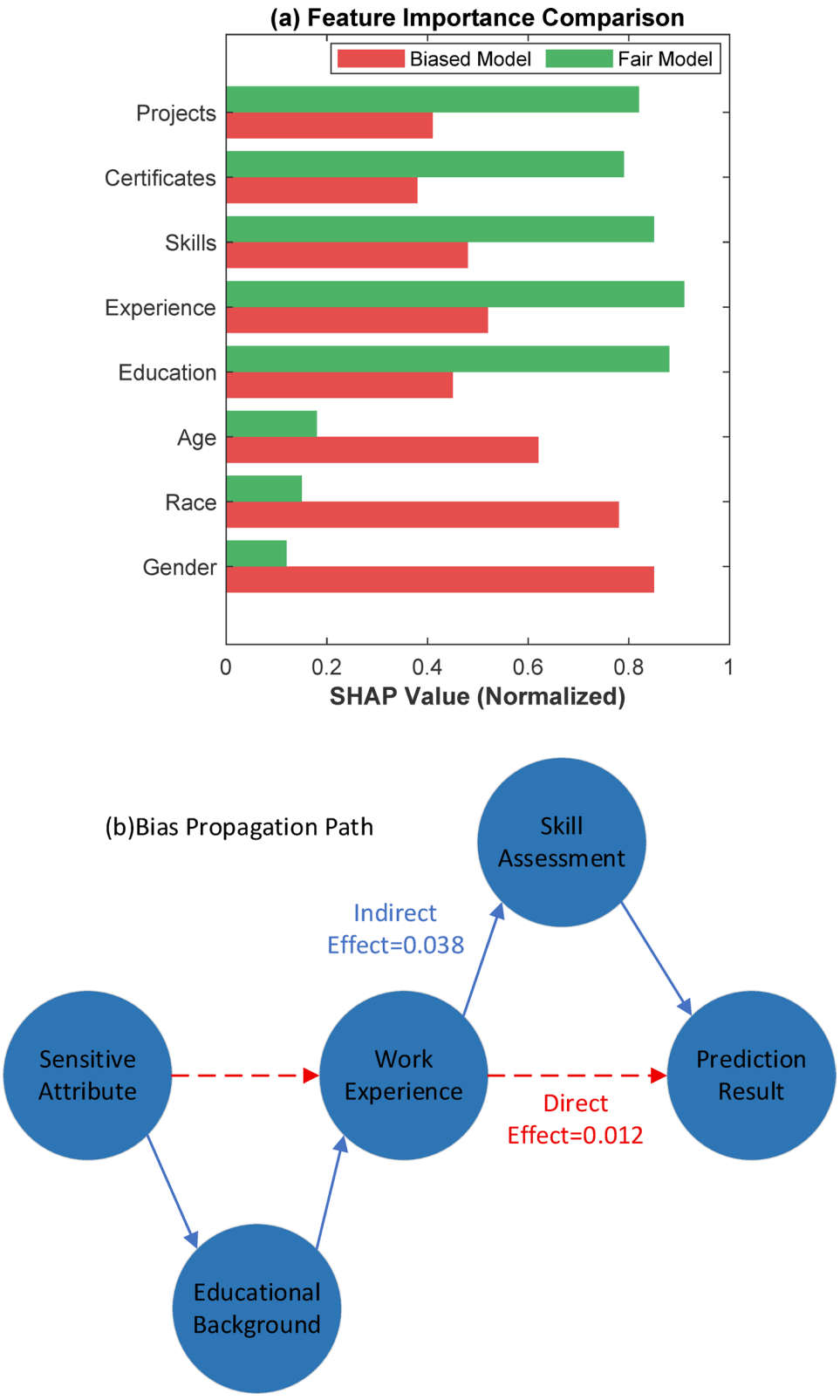


Fig. 6. Bias source and propagation mechanism visualization analysis. (**a**) Feature importance comparison; (**b**) Bias propagation path; (**c**) Counterfactual fairness analysis.

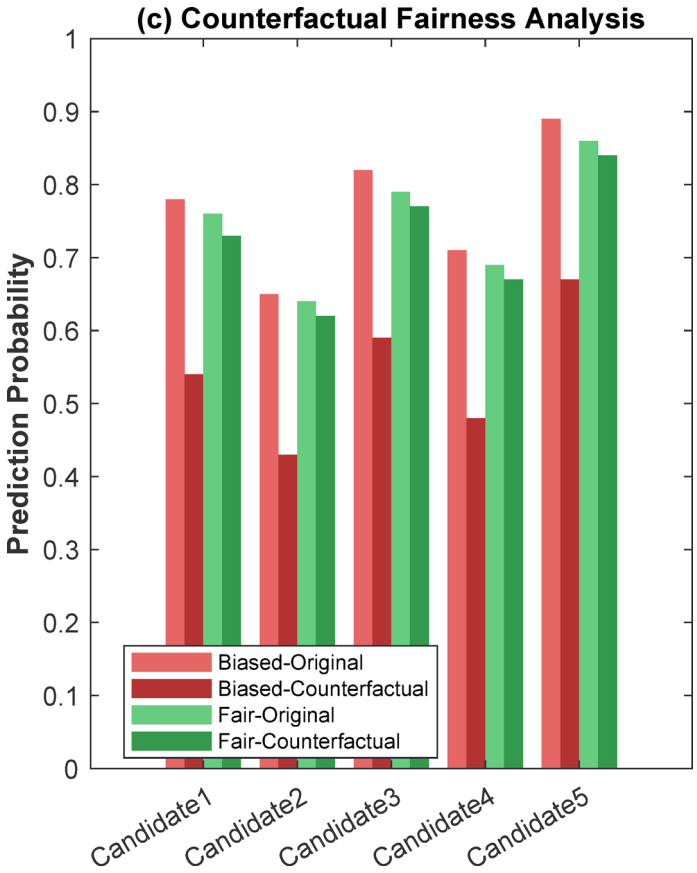


Fig. 6. (continued)

of prediction accuracy while significantly improving fairness measures. This innovation further confirms the application value of multi-task learning in algorithmic fairness. The construction of a multi-dimensional fairness evaluation system is another major contribution to this work.

By combining nine measures in three broad classes of group fairness, individual fairness, and causal fairness, this system evaluates the overall fairness performance of hiring systems. Experimental results demonstrate the existence of trade-off relationships among various measures of fairness as advocated in the mathematical incompatibility of fairness definitions theory by Loi, Michele et al.[12]. Adhering to the framework of the AI Fairness 360 toolkit by Bellamy et al.[35], the paper rigorously proves such trade-off relations based on Pareto frontier analysis techniques and sets up quantitative grounds for firms to choose effective fairness metrics within real-world applications. It must be noted that equal opportunity and rate parity are generally considered more in hiring because the former is more aligned with merit-based selection criteria. Through the addition of interpretability analysis module, it becomes possible to identify the source of bias. Following Wang, Huanjing, et al.'s[36] SHAP value analysis technique, this platform does not only detect influential features leading to bias but also follows the paths of bias propagation through the model.

Experimental analysis confirms that attributes that appear insensitive like postal codes, educational history, and work experience can serve as proxy variables for sensitive traits, as supported by the study of Ullah Azmat, et al. on online advertisement platforms[18]. Using gradient-weighted class activation mapping and causal mediation analysis, this platform identifies the precise sequences and attributes that introduce bias with ease, and it supplies actionable feedback towards implementing targeted bias reduction strategies.

Limitations and threats to validity

Though our approach shows encouraging results, we would like to highlight a few key points regarding our results and how our approach can be used in practical implementations.

To begin with, our validation study relies solely on Western datasets (US and Europe; Table 2), which belong largely to technology-related sectors. A dangerous level of risk is involved in this data geographical and categorical bias. Cultural factors concerning protected norms and laws regarding equality vary extremely widely—other countries emphasize distinct variables such as caste (India), language (Belgium), and political party membership. One needs to be extremely cautious when our measure for bias validation relies for calibration purposes on Western values such as demographic equality and equality of opportunity. Otherwise, we would be unable to argue that our revealed bias would amount to much in other parts of the globe: Asia, Africa, Latin America, the Middle East. Also, we might be dealing with a technology-related bias model that does not apply to other sectors such as healthcare, manufacturing, education, public administration.

Method	Training time	Peak memory	Estimated cloud cost*
Random forest	2–3 h	8 GB	$5–8
MLP	4–6 h	12 GB	$10–15
Adversarial	10–14 h	22 GB	$25–35
Fair representation	8–12 h	18 GB	$20–30
Proposed method	18–24 h	32 GB	$50–80

Table 6. Computational cost comparison (Bias in Bios Dataset, 397,340 samples). * Based on AWS p3.2xlarge instance pricing (~$3/hour).

Secondly, our class imbalance ratio values (3.2:1 to 4.1:1) are relatively moderate compared to those found in practical problems. In fact, for jobs such as recruiting executives and/or for more technical jobs, class imbalance ratios may be much larger—ratios of 20:1 or 100:1 are possible. In such extreme class imbalances, our adversarial training process and our fairness measures can become erratic: our model may be learning to be generally accurate while predicting little for minority groups. Additionally, our class measures may become numerically unstable. We have made no effort to test our model for such conditions. This becomes a serious issue for minority groups who face class imbalances in their sensitive attribute.

Beyond geographic and sectoral constraints, our use of curated public datasets does not reflect the data quality challenges of real-world Applicant Tracking Systems (ATS). Research datasets like Adult, Bias in Bios, and Kaggle Resume are relatively clean and structured, whereas production ATS data typically exhibits substantial missing values (30–50% incomplete profiles), inconsistent formatting, and unstructured text requiring natural language processing. Our framework assumes preprocessed, feature-engineered inputs and does not address: (1) missing data handling without disadvantaging affected groups, (2) fair representation learning from raw, heterogeneous resume formats (PDFs with varied layouts), or (3) maintaining fairness when data completeness varies systematically across demographics. Real ATS systems also process multimodal data—video interviews, unstructured cover letters, portfolio links—that our text-and-tabular framework does not accommodate. Robust deployment would require augmenting our architecture with missing data imputation, automated feature extraction from unstructured text, and fairness validation across varying data quality levels.

Thirdly, computational requirements preclude accessibility. Our model needs strong GPU machines (NVIDIA A100 with 40GB GPU memory requirements), and lots of data would be preferred. Training our model took 18–24 h with a 32GB peak memory usage in our experiments for the Biased in Bios dataset.

Table 6 provides a comparative analysis of computational requirements across methods. Our proposed framework requires 4–6× longer training time than simpler baselines, with proportionally higher memory and cost requirements. This computational overhead represents the trade-off for achieving 12–18% points improvement in intersectional bias detection and comprehensive fairness guarantees.

While the computational demands are substantial, they remain feasible for organizations with moderate IT budgets, particularly considering that model training is typically performed offline and infrequently (e.g., monthly or quarterly retraining cycles).

This could be a major hindrance for Small and Medium-Sized Enterprises (SMEs), which may have fewer IT developments. The irony remains that those regions and organizations who might be most affected regarding bias problems would be least equipped to use our solution. Also, for smaller datasets (with 500-1,000 samples), our solution's performance characteristic remains unexplored.

Fourth, even though we offer nine measures of fairness for comparison across three dimensions, our design does not address the underlying challenge of making good decisions about which measures of fairness are most appropriate for a particular use. While our overall measure of fairness relies on weights that are arbitrary ($w_g = 0.4$, $w_i = 0.3$, $w_c = 0.3$), this does not address questions about values. Where we prioritize fairness depends on our values. And our values may require us to prioritize measures differently based on legal mandates and our particular contexts. There are underlying trade-offs among particular measures: for example, increasing one measure of demographic parity may decrease another measure of predictive rate parity.

Fifth, our approach does not consider longitudinal studies of temporal dynamics and effects of feedback loops. We measure fairness based on static datasets for a snapshot moment in time. By contrast, a realistic use of AI for recruiting would be a real-time process. As pointed out in work done by Bauer et al.[21], biased decisions lead to a creation of a feedback loop where retraining algorithms increase bias levels. We cannot tell whether our results are stable for several model rounds or whether a drift occurs with a changing composition of applicants.

Lastly, our assessment solely targets detection quality and fairness values but does not validate other elements of practical deployability: user acceptance among HR professionals in this process, bias mitigation recommendation actionability, legal adequacy according to current regulatory changes (such as EU AI Act), and cost-value for recommended bias mitigation. Without field and deployment studies, we cannot verify effective deployability of our technical progress.

Such limitations are extensive in restricting our external validity. We would recommend any organization thinking of applying our framework to perform a pilot test with their data before applying our model. We would also recommend that our model be considered more of a foundation for other methods of research and therefore be adapted and validated.

Legal, regulatory, and ethical considerations

Applying our framework in a practical recruitment scenario needs to be done while considering legal, regulatory, and ethical issues that go beyond technical considerations of fairness.

Legally speaking, our work engages with anti-discrimination laws that exist differently in various jurisdictions. Under United States law, for example, Title VII of the Civil Rights Act prohibits workplace discrimination and requires employers to comply with the "disparate impact" rule, where they must prove their selection criteria are job-related and justified when composed of groups protected by law. Although our disparate impact measure (DI) complies with the "80% rule" employed by the United States Equal Employment Opportunity Commission, European Union laws regulate our requirements more stringently via General Data Protection Regulation (GDPR) laws and the pending AI Act in prohibits our transparency requirements and certain automated decisions. This is because legal sufficiency does not necessarily equate to mathematical sufficiency.

Handling data privacy and consent requires additional consideration. While our scheme necessitates access to and processing of sensitive data (gender, race, and age) for bias detection purposes, data protected by Article 9 of the GDPR is considered a special category that necessitates consent and a legitimate interest for processing. Additionally, data such as demographic information may be considered unlawful to collect in some countries. This requires privacy-preserving methods for data access and consent.

Alignment with Stakeholders: Alignment with stakeholder values introduces normative issues that our technical infrastructure does not address. Multiple stakeholders such as job applicants, HR staff, hiring managers, attorneys, and diversity officers may have varying levels of emphasis for different types of fairness. Individual fairness might be more important for applicants than for diversity officers. Our equality weights (Eq. 17) are a reflection of our values as researchers. They might vary for other stakeholders.

Thirdly, issues concerning potential errors of over-correction and under-correction should be taken into consideration. Over-correction could be a potential cause for reverse discrimination charges and/or may prefer less qualified applicants. On the other hand, failure to correct may continue to propagate inherent bias. While our approach supplies detection and methods for assessment, evaluation regarding appropriate levels of correction may vary in accordance with values and circumstances.

In other words, technical fairness alone does not constitute responsible deployment. This requires action on the part of organizations to involve legal counsel and ethics boards in converting our capabilities into legal and moral best practices for recruiting.

Future directions

Several promising directions warrant further investigation. First, extending from bias detection to automated bias mitigation would enable end-to-end fair AI development—automatically adjusting model parameters or reweighting training data upon detecting bias patterns. Second, addressing temporal dynamics through longitudinal studies could characterize how fairness guarantees evolve under continuous retraining and feedback loops, informing strategies for maintaining long-term fairness. Third, enhancing robustness to real-world data challenges—handling missing values, extracting features from unstructured text, and processing multimodal inputs—would bridge the gap between research datasets and production ATS deployments. Fourth, developing computationally efficient variants through model compression or federated learning could democratize access for resource-constrained organizations. Fifth, cross-cultural validation on non-Western datasets (Asia, Africa, Latin America) and non-tech sectors (healthcare, manufacturing, education) would establish broader applicability. Finally, the methodological framework integrating multi-task learning and causal inference could be adapted to other high-stakes domains including loan approval, criminal justice risk assessment, and medical diagnosis, advancing responsible AI across critical decision-making systems.

Conclusion

In order to deal with the algorithmic biases of AI hiring systems, this research suggests a deep learning-based bias detection model, making a contribution to a new avenue of algorithmic fairness research and practice. The key contributions of this study are three-fold: Firstly, a multi-task adversarial learning framework is developed, which realizes fine-grained intersectional bias detection via attention mechanisms, with 12–18% points improvement in detection accuracy; secondly, a multi-dimensional fairness assessment system is established, which encompasses nine metrics within three broad categories of group fairness, individual fairness, and causal fairness, with Pareto frontier analysis demonstrating the trade-off relationships between various fairness metrics; thirdly, an interpretability module incorporating SHAP values, gradient-weighted class activation mapping, and causal mediation analysis is embedded, realizing accurate localization and visualization of the sources of bias. Experimental findings confirm the efficacy of the suggested approach in bias detection accuracy, completeness of fairness analysis, and result interpretability. Future work can be done in the following directions: First, realize automated bias mitigation methods to enable end-to-end fairness optimization from bias detection, forming a complete closed loop for fair AI systems; second, increase the range of applications of the framework by applying the methodology to other high-stakes decision-making domains like loan approval, medical diagnosis, and college admission, examining the framework's strength and generalizability; third, explore dynamic mechanisms of bias evolution, examine the long-term effect of bias under feedback loops, and develop fairness assurance mechanisms with the time factor taken into consideration. Through these efforts, this research will contribute to building more fair, transparent, and responsible artificial intelligence systems.

Data availability

The datasets used in this study are publicly available. The Adult Census Income dataset is accessible at https://archive.ics.uci.edu/ml/datasets/adult. The Bias in Bios dataset is available at https://github.com/microsoft/biosbias.

The Kaggle Resume Dataset is available at https://www.kaggle.com/datasets/gauravduttakiit/resume-dataset. All three datasets, along with the complete implementation code for this study, have been integrated and are jointly accessible at https://github.com/neththomassoliz/bias-detection-framework (archived at https://doi.org/10.5281/zenodo.19382457). No additional data were generated in this study. Researchers wishing to request further information should contact the corresponding author, Yujiao Du, at qopp1812@outlook.com.

Code availability

The custom code implementing the proposed framework is publicly available at https://github.com/neththomassoliz/bias-detection-framework (archived at https://doi.org/10.5281/zenodo.19382457). There are no restrictions on access. For implementation inquiries, please contact the corresponding author, Yujiao Du, at *qopp1812@outlook.com*.

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Conceptualization, J.W. and Y.D.; methodology, Y.X. and R.L.; formal analysis, J.W. and Y.X.; investigation, all authors; data curation, R.L. and Y.D.; writing—original draft, J.W. and Y.D.; writing—review and editing, Y.X. and R.L.; visualization, Y.X.; supervision, Y.D.; project administration, J.W. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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